Prince Verma

M.Tech Smart Mobility - TA

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Degree	University/Institute	Year	CGPA/ Marks(%)
M.Tech Smart Mobility - TA	IIT Hyderabad	2026	7.85
B.Tech	Babu Banarasi Das National Institute of Technology and Managaement,Lucknow	2022	8.6
XII (CBSE)	Rani Laxmi Bai	2018	86.4%
X (CBSE)	Rani Laxmi Bai	2016	10

RESEARCH/PUBLICATION

WildSpoof Challenge — ICASSP 2026 Grand Challenge (IEEE ICASSP Conference)

- Built a Spoofing-Aware Speaker Verification (SASV) system for the WildSpoof Challenge 2025, which must reject both wrong speakers and fake (synthesized) speech recorded in real-world noisy conditions.
- Used the SpoofCeleb dataset, which contains over 2.5 million utterances from 1,251 speakers, paired with fake speech generated by 23 modern text-to-speech (TTS) systems.
- Studied the challenge baseline (SKA-TDNN) and the a-DCF metric, which jointly penalizes accepting a wrong speaker and accepting fake speech.
- Designed a trial-level scoring pipeline for target vs. non-target vs. spoof detection, and submitted continuous likelihood scores for each trial following the challenge protocol.

THESIS

Guide :- Sri Rama Murty Kodukula
Topic:- Audio Deepfake Detection using Cross-Attention Extended SLIM (SLIM-XA)

- Carried out the research in four phases: speaker identification with x-vectors, speaker verification with CA-MHFA, a CNN-based deepfake detector, and finally the proposed SLIM-XA system.
- Built a CNN deepfake detector using mel-spectrogram, MFCC, zero-crossing rate, and spectral centroid features, reaching about 95% accuracy on seen attacks but only about 67% on unseen attacks, which showed that artifact-based detectors do not generalize.
- Deployed the CNN detector as a Flask web app that accepts WAV, MP3, FLAC, OGG, and M4A files (up to 50 MB) and shows the waveform, mel-spectrogram, confidence score, and a plain-language explanation of the prediction.
- Extended the SLIM framework (NeurIPS 2024) by replacing its fixed distance measure with a bidirectional 4-head cross-attention module that predicts style features (emotion-tuned Wav2Vec2-XLSR, layers 0–10) from linguistic features (ASR-tuned Wav2Vec2-XLSR, layers 14–21) and vice versa; the prediction errors act as the signal for spotting fake audio.
- Built a two-stage pipeline: Stage 1 trained on ~6,890 real speech samples only (ASVspoof 2019 + MILABS) using alignment, orthogonality, and cross-attention residual losses to learn how style and content naturally match; Stage 2 trained a real/fake classifier on a 1536-dimensional, six-stream feature set (up from SLIM's 1024) with the encoders, compression, and attention modules kept frozen.
- Implemented a bottleneck compression module and reproduced the original SLIM loss with batch normalization and $\lambda = 0.007$.
- Tested on the ASVspoof 2019 LA evaluation set containing 13 unseen attack types, achieving 99.98% fake-speech recall, 73.2% accuracy, and 22.37% EER, while training on only 50% of the data on a single 11 GB NVIDIA Titan Xp GPU (about 1 hour for Stage 1 and 5.5 hours for Stage 2).

WORK EXPERIENCE

ML Engineer Intern — Computer Vision | Coral Innovation Pvt. Ltd.

Civic Issue Detection

- Architected and trained a high-performance YOLOv8m detector for 10-class civic-issue recognition on a 17,000+ image dataset, driving test performance to 0.78 mAP@50 / 0.52 mAP@50-95 through systematic hyperparameter optimization, transfer learning, and early-stopping — delivering state-of-the-art accuracy on a challenging real-world, long-tailed detection task.
- Owned the end-to-end data strategy that unlocked the model's accuracy gains, consolidating and QA-auditing a noisy raw corpus, collapsing 7 confusable animal classes into one robust category, and quarantining ~4,500 mislabeled samples — directly lifting per-class precision/recall and slashing false positives.
- Pioneered a domain-robustness augmentation pipeline using synthetic night/rain/fog transformations to harden the model against adverse field conditions, dramatically improving generalization on low-light and weather-degraded imagery where baseline models failed.
- Shipped the model to production on edge hardware, compressing the network to an INT8 TFLite target (~26 MB) with NNAPI acceleration and rigorously benchmarking the accuracy–latency frontier to achieve real-time, fully-offline on-device inference under tight mobile compute and memory budgets.

Meter Reading Automation

- Defined quality metrics for a meter-symbol detection system and compared fine-tuned YOLO variants on held-out validation sets, measuring accuracy and reliability under different lighting conditions.
- Studied OCR errors on meter serial numbers, traced them to specific problem inputs, and built a preprocessing step (denoising, deskewing, cropping) that filtered out low-quality images before inference and improved recognition accuracy.
- Combined detection and OCR into one automated pipeline, delivered with the engineering team through Agile sprints, which reduced manual data entry and increased throughput.

Internship — Transmission Line Fault Detection | Power Grid Corporation of India Limited (PGCIL) | Jan 2024 – May 2024

- Used drone-captured imagery to identify and analyze faults in transmission lines.

PROJECTS

Traffic Sign Classification Using CNN for Autonomous Vehicles

- Developed a deep learning system for traffic sign classification, achieving 99.5% accuracy on the German Traffic Sign Recognition Benchmark (GTSRB).
- Designed a custom CNN that recognizes 43 traffic sign categories in real time, supporting autonomous vehicle perception and advanced driver assistance systems (ADAS).

Searchable Knowledge Base from PDFs using Retrieval-Augmented Generation (RAG)

- Built a local RAG pipeline using LangChain, HuggingFace embeddings, and FAISS to answer questions across multiple PDF documents using Groq LLMs.
- Designed an end-to-end workflow for loading, chunking, embedding, and searching PDFs, allowing natural-language questions across many documents.
- Used vector similarity search with large language models to retrieve relevant content and generate accurate, context-aware answers.

AgriTech AI Analyzer — Agricultural Monitoring System

- Developed a Flask-based web application that uses AI to help farmers monitor and manage their crops.
- Integrated multiple ML models (YOLOv8, EfficientNet, Random Forest) for crop, soil, pest, and disease analysis, plus nutrient analysis and precision farming, across six agricultural areas.
- Achieved about 94% accuracy, added Grad-CAM visualizations to show which image regions drove each prediction, and deployed the app with Docker for real-time use.

SCHOLASTIC ACHIEVEMENTS

- **Cleared GATE 2024** with an **All India Rank of 498**.
- **Winner**, *AgriTech AI Hackathon 2025* (KIT Skill Hub & Pavaman Technologies) – Developed **AgriTech AI Analyzer**, a Flask-based AI system for **crop, soil, pest, and disease analysis** using **YOLOv8, EfficientNet, and Random Forest** models; achieved **~94% accuracy**, integrated **Grad-CAM visualizations**, and deployed via **Docker** for real-time, user-friendly inference.

POSITIONS OF RESPONSIBILITY

- **Head, Volunteer Coordinating Unit (VCU)**—Prayas Club, IIT Hyderabad
- **Coordinator, Alumni Relation Cell** – Ideation and Operations Coordinators

RELEVANT COURSES

Foundations of Machine Learning | Advanced Computer Networks | Internet of Things (IoT) | Random Variables and Stochastic Processes | Deep Learning | Multimedia Content Analysis |Autonomous Navigation | Automotive Communication and Sensing| Traffic Engineering & Intelligent Transportation Systems